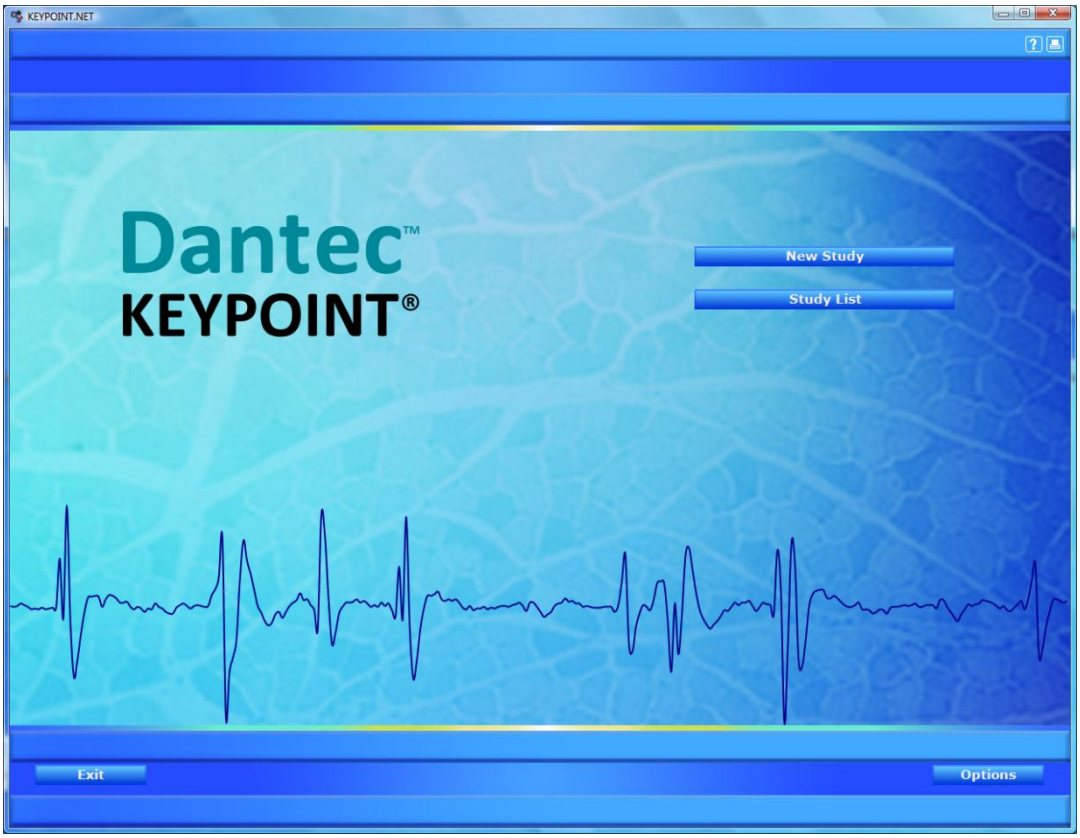
natus neurology

Dantec™ KEYPOINT®



Software Instructions for Use

! USA Rx only

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At the time of printing/transfer to the DVD, this manual correctly described the device and its functions. However, as modifications may have been carried out since the production of this manual, the system package may contain one or more addenda to the manual. This manual including any such addenda must be thoroughly read, before using the device.

The following situation voids any guarantee(s) and obligations for Alpine Biomed ApS:

- The device is not used according to the enclosed manuals and other accompanying documentation.

This system is CE marked in conformity with the requirements in the Medical Device Directive 93/42/EEC.

Alpine Biomed and Keypoint are registered trademarks of Alpine Biomed in the U.S. and in other countries.

Note that the documentation accompanying the Keypoint devices include~~s~~ a Software Instructions for Use manual, a Hardware Instructions for Use manual, a Technical Data Sheet and a Clinical Reference Manual. Both the Software and Hardware Instructions for Use manuals are necessary to be able to use the Keypoint.NET application and carry out a test. For advanced use of the Keypoint.NET application, the Clinical Reference Manual (available in English only) offers a description of all the functions and tests that can be performed. For technical information, the Technical Data Sheet (available in English only) offers detailed technical specifications.

Consult Instructions for Use / Attention to Warnings and Cautions

Before starting a test, make sure you read the documentation and fully understand how to operate the system. Pay special attention to all Warnings and Cautions.

For the list of the general regulatory symbols associated with this device, see the General Regulatory Symbols section in this manual.

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Safety Information

The system is intended to be used in electrophysiological tests such as: Electromyography (EMG), Nerve Conduction Studies (NCS), and Evoked Potential (EP) recordings.

Signs and Symbols

Sign/Symbol	Explanation
! USA Rx only	For US audiences only. US laws restrict this device to sale by or on the order of the physician.
	Warnings / Precautions associated with this device.
	Consult Instructions for Use.
CE 0543	The device complies with the EC directive 93/42/EEC on medical devices.
Hz	Repetition Rate
	Repetitive Stimulation
	Single Stimulus
µs	Stimulus Duration
	Stimulus Intensity Mode Indicator
	Stimulus Release Indicator
	Sweep Speed / Sensitivity
	Wheel Mode Indicator (Intensity)
	Wheel Mode Indicator (Marker/Trigger)

Intended Use

Keypoint.NET is intended as an electrophysiological aid to assess diagnosis and prognosis, and to monitor diseases of the central and peripheral nervous system. It can also be used to study functional aspects of nerves and muscles in other fields such as rehabilitation (physical medicine), occupational medicine, and sports medicine.

CONTRAINDICATIONS

It is a medical decision, whether the risk of use clearly outweighs any possible benefit in each case.

A patient with an implanted electronic device (e.g., a cardiac pacemaker) should not be subject to electrical stimulation unless medical specialist opinion has first been obtained.

In conditions with bleeding tendency, certain care should be taken when needles are used.

Conventional precautions should be taken with patients with infectious diseases.

Age and sex do not in themselves present contraindications for any procedure.

For contraindications with needles / electrodes, refer to their accompanying instructions for use.

WARNINGS

For warnings and precautions/cautions related to the hardware and its use, refer to the hardware manual. Make sure to read the information before proceeding to use the equipment.

Do not use this PC-based equipment for anything else than it is intended for by the manufacturer, i.e. carrying out tests on patients and possibly subsequent report generation. Do not install any other software than the Keypoint.NET Software. Alpine Biomed ApS assumes no responsibility when not used as described in this manual.

Do not touch the wheel mode button, while moving the cursors, or while changing the trigger levels, using the intensity wheel, as you might unintentionally increase the stimulation intensity.

When operating the current stimulators, be careful not to expose patients to high currents. Therefore, before connecting, or disconnecting the stimulation electrode, always “reset” the stimulator.

Pay attention to the intensity indicator during the use of the program. −See the section Stimulator Overload in this user guide, and the section Stimulators in the Keypoint hardware manual for further information.

Avoid transthoracic stimulation.

Should a skin rash or any other unusual symptoms develop during use, stop stimulation and remove electrode from skin.

PRECAUTIONS

Avoid electrical stimulation for an extended period of time.

Avoid accidental contact between connected −even when not applied on the patient− electrodes and other conductive parts, including those connected to protective earth.

In case of bleeding from needle electrode insertions, protective wear should be used according to local regulations.

Conventional precautions should be taken with patients with infectious diseases.

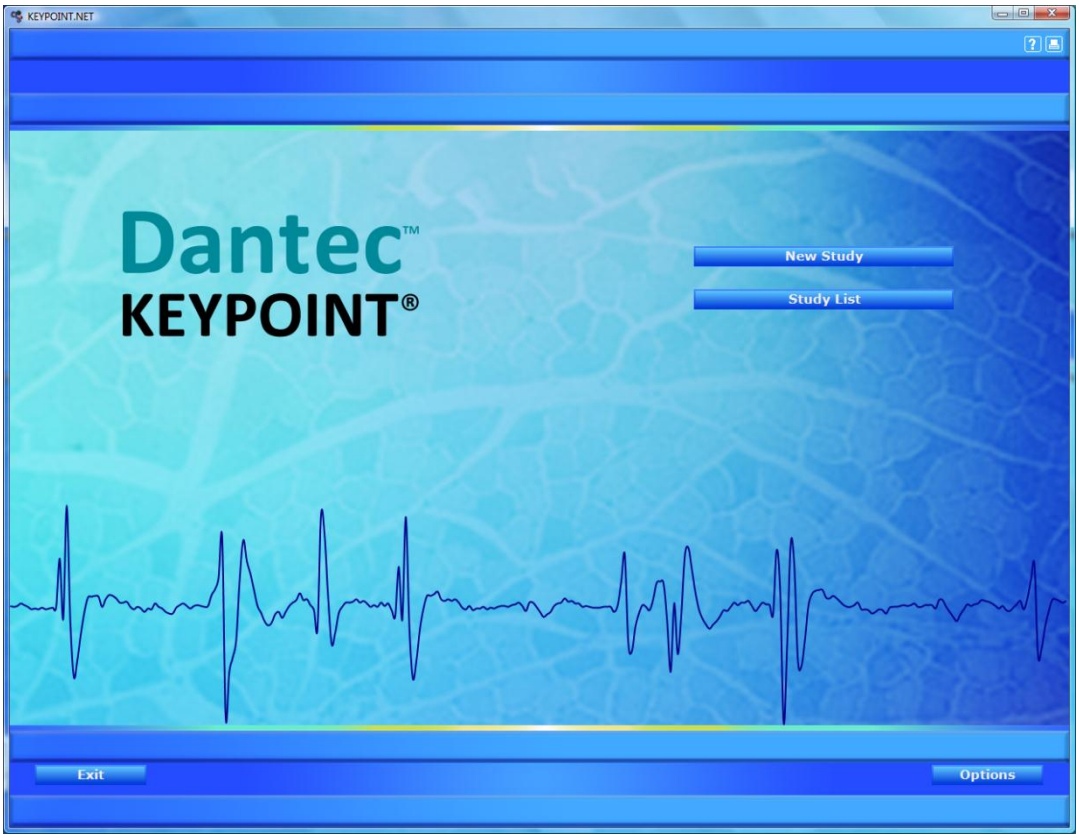
ADVERSE REACTIONS

No adverse reactions other than bleeding and transient pain due to needle electrode insertion have been reported.

Starting Keypoint.NET

To start the application, double-click the Keypoint.NET icon located on your Windows desktop. The ***Home Page*** will then be displayed.

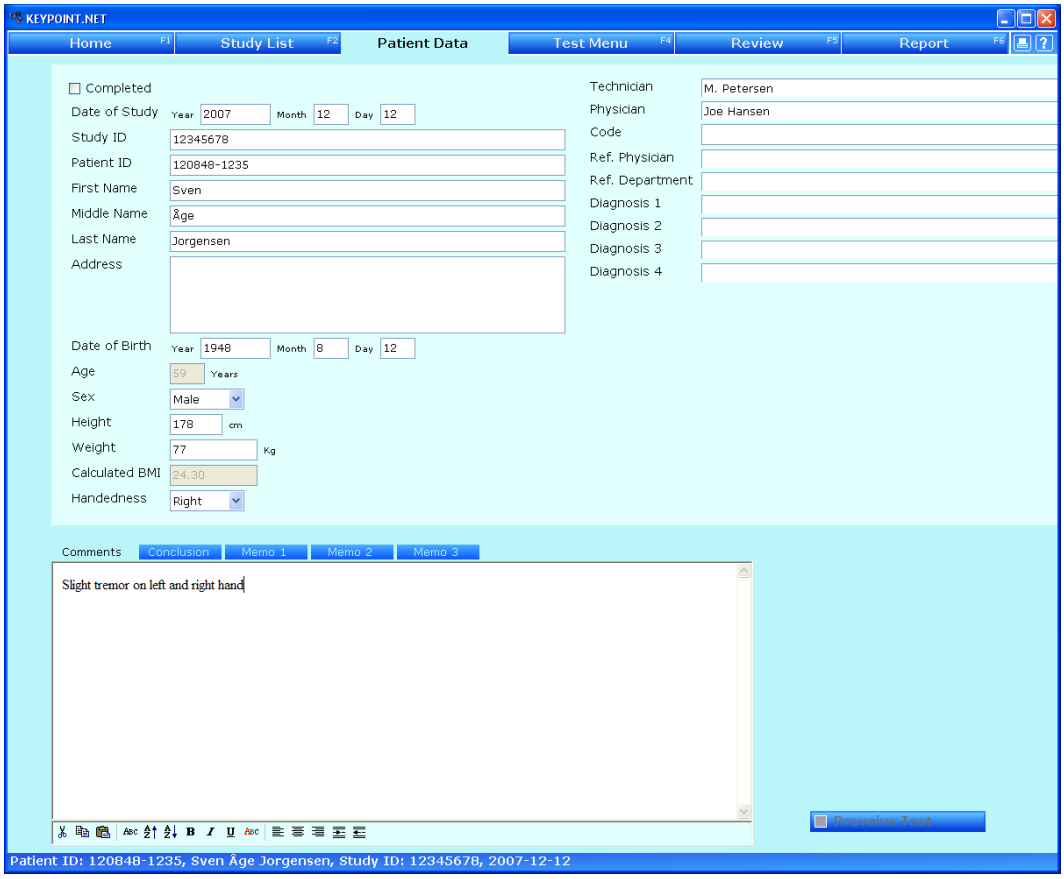
Home Page



The ***Home Page*** is the start view when the program opens. From the ***Home Page*** the following options are available:

- Create a ***New Study***.
- Open the ***Study List*** with existing studies.
- Access the advanced ***Options*** −only accessible from the home page.
- Exit the program.

Patient Data



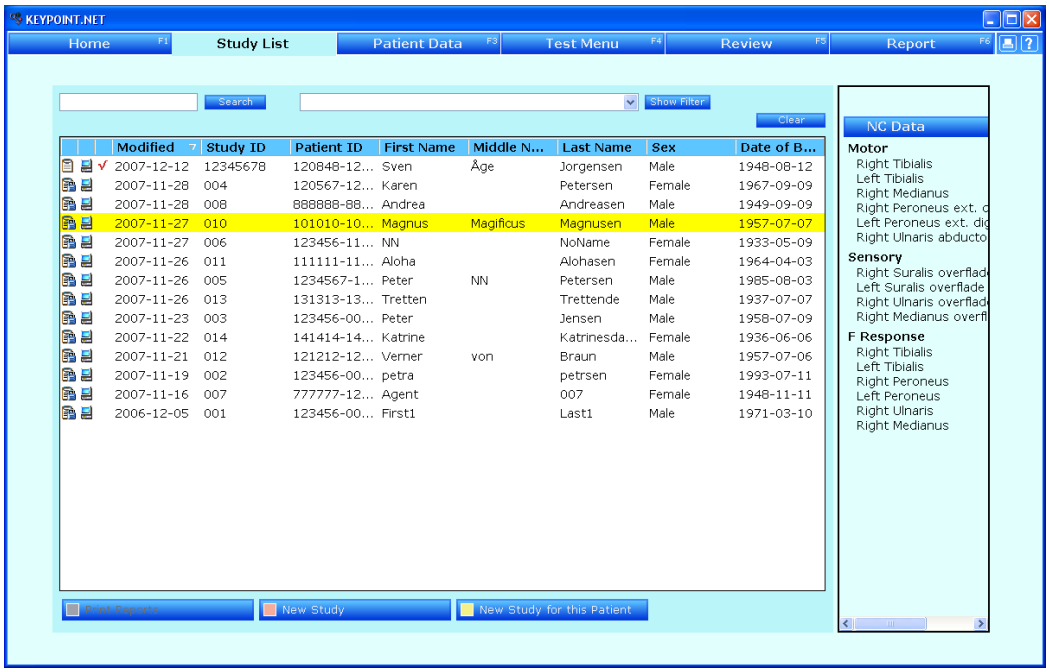
Clicking the ***New Study*** button in the Home page opens the ***Patient Data*** page.

From the Patient Data page, you can enter/edit:

- Patient data
- Comments
- Conclusions

Note that data cannot be entered when the ***Study Complete*** check box is selected.

Study List

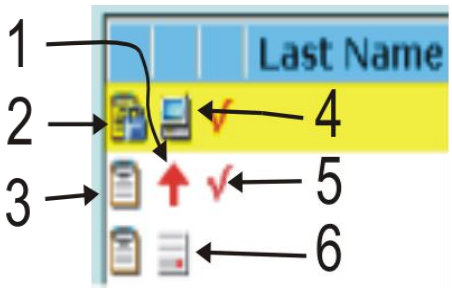


Clicking the ***Study List*** button in the Home page, or the ***Study List*** tab on the Navigation Bar, opens the study list, from which you can access a study for editing.

The study list may consist of both local and remote studies.

Clicking a study highlights the relevant study and provides a "review list" in the *right* side of the study list view, as shown in the figure above.

Status Icon



1 Uploading	**4** Local
2 Completed	**5** Checked out
3 Active	**6** Remote

Double-clicking a Study

Double-clicking a study opens the ***Patient Data Menu*** tab for editing the data entry fields.

New Study for this Patient

To create a new study for the patient, click ***New Study for This Patient***, and you are directed to the ***Patient Data*** page.

Right-clicking a Study

Right-clicking a study displays a shortcut menu that allows you to export, delete, and generate reports. You can also sort, and filter the studies.

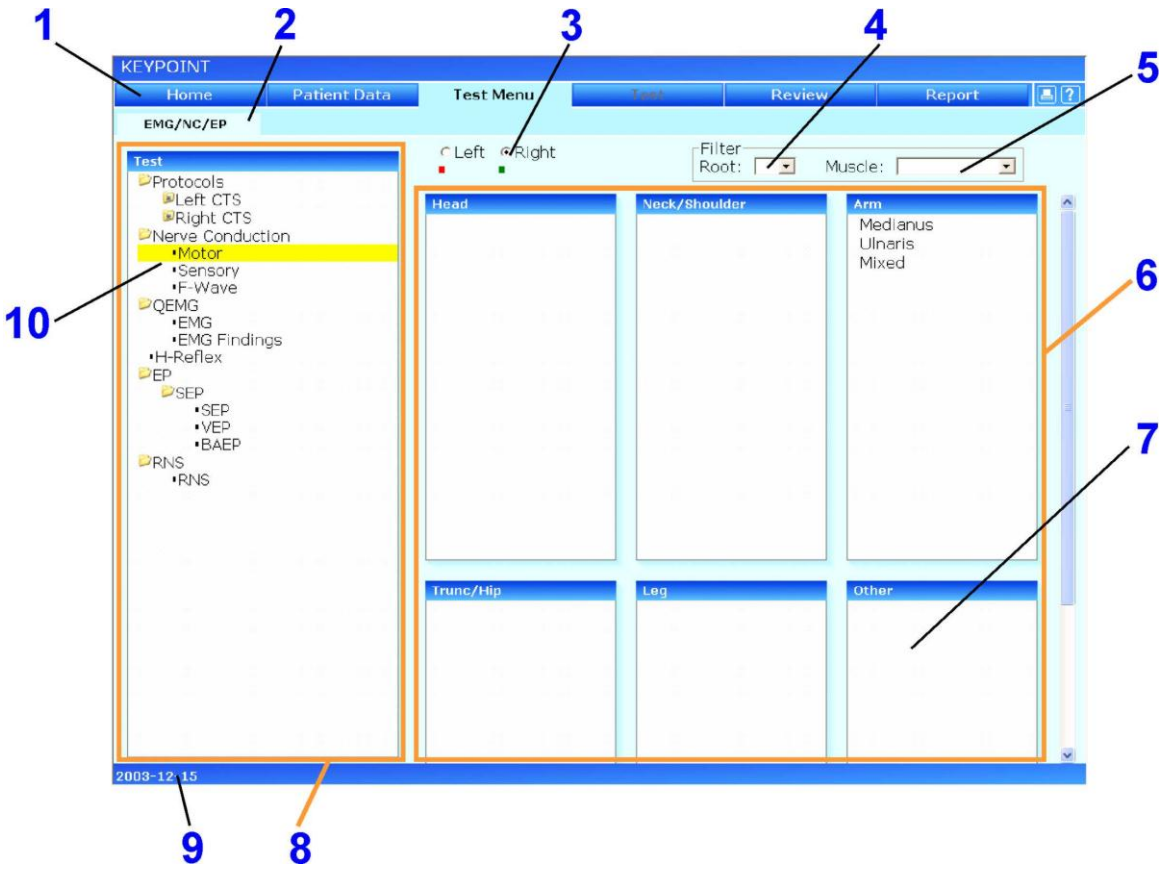
Right-clicking a Column Heading

Right-clicking a column heading, displays a list of column types that you can select to be displayed in the study list.

Left-clicking a Column Heading

Left-clicking a column heading, sorts the list according to the column.

Test Menu



1 Top navigation bar with tabs
2 Application Setting Selection
3 Left, or right option buttons
4 Filter root, drop-down list
5 Muscle, dropdown list
6 Detailed sites view
7 Anatomy box
8 Test type view
9 Patient status bar
10 Tree view structure

Test Types and Detailed Site Views

You select a test from the ***Test Menu*** tab, which can be accessed from the ***Patient Data*** tab. Clicking the test menu tab displays the ***Test Types*** view (**8**) and the ***Detailed Sites*** view (**6**). The test type view consists of test folders. Each folder provides organization of various related tests. The detailed sites view displays a number of ***Anatomy Boxes*** (**7**) containing the available muscle / nerve sites. –See figure above.

Starting a Test

1. To select a test, click the test, e.g. ***Motor***, from the tree view.

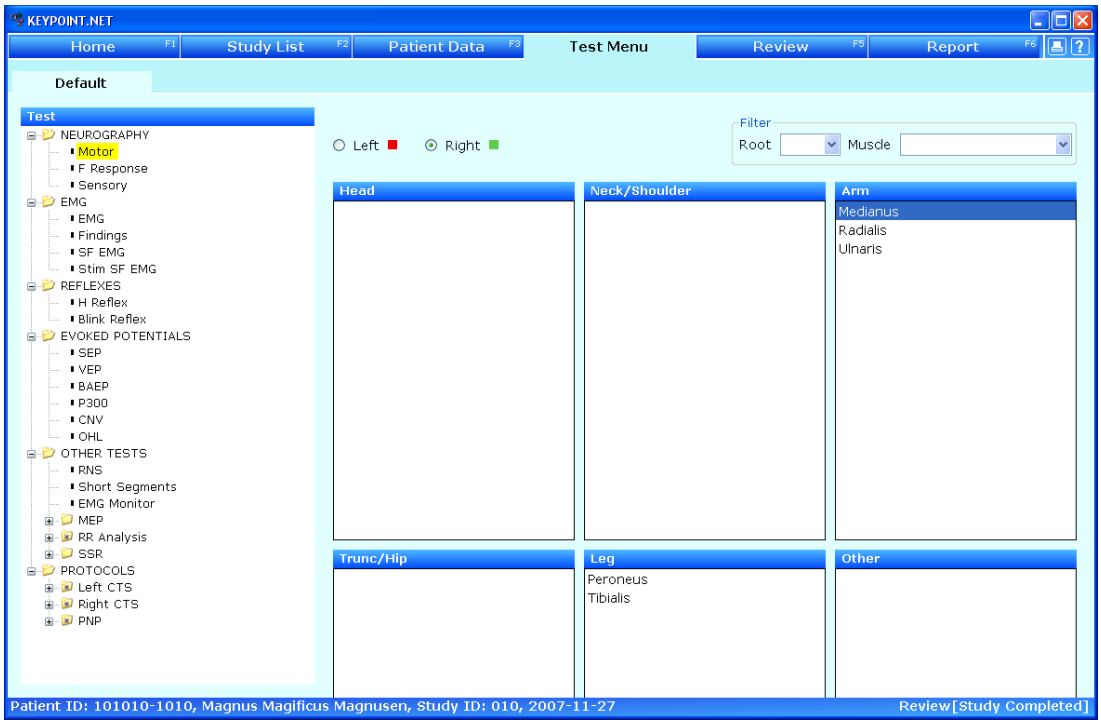
   –The right-hand joystick/arrow keys of the Control Panel can also be used for selecting tests.

2. Then select the side – ***Left***, or ***Right*** – by clicking the appropriate option.

   –On the Control Panel, the orange and green buttons can also be used.

   –Note that you can choose the ***Filter Root*** and the ***Muscle /Nerve*** from the list provided in the drop-down boxes. Click the drop-down arrow and the list opens as shown in the figure below.

3. Finally, click a stimulation site from the ***Detailed Sites*** view, e.g. Medianus, in the ***Anatomy box***. This action automatically directs you to the ***Test Menu*** page.



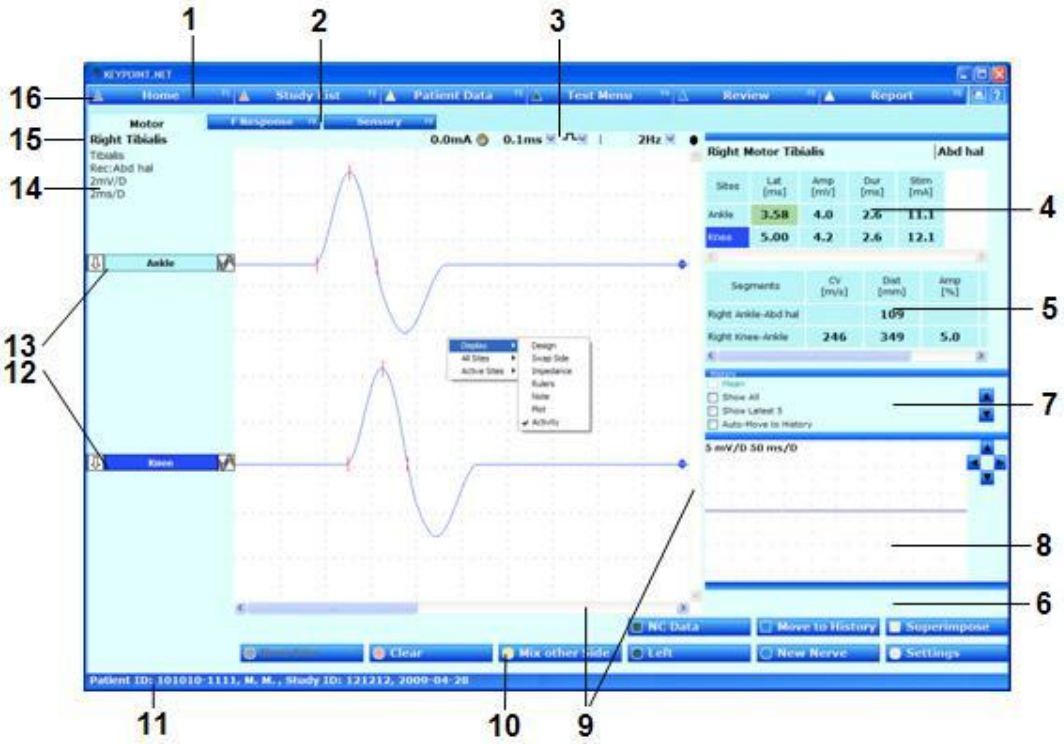
Test Example 1: Motor Nerve Conduction

Introduction

The Keypoint.NET allows you to perform a series of tests on nerve conduction. These tests are listed on the Test Menu page. As an example, the Motor Nerve Conduction Test functions are described below.

Note that all the other tests have similar functions and are conducted in a similar way.

Overview – Motor



1 Top navigation bar with tabs

2 Test navigation tabs

3 Stimulation settings bar

4 Site table with scroll bar

5 Segment table with scroll bar

6 Optional windows (provided by the right-click menu)

7 History window

8 Optional activity window

9 Scope window with horizontal and vertical Scroll bars

10 Color marked buttons (for use with the dedicated keys)

11 Patient status bar

12 Selected sites

13 Deselected sites

14 Trace information area

15 Name and site of the anatomy chosen

16 Using the Control Panel – G4 / Focus

Notes on Functions and Views

Test Navigation Tabs (2)

Related tests, which are grouped, are placed in separate ***Test Navigation*** tabs. These can be selected using the PC Keyboard keys F7 – F12, and by clicking with the mouse.

On the Keypoint G4/FOCUS, the ***Test Navigation*** tabs can also be selected using the ***Software Navigation*** Right and Left keys ◄ ► .

Patient Status Bar (11)

The ***Patient Status Bar*** provides the following information: Patient ID, Patient Name, Study ID, Date, and Test Status.

Scope Window

Working with Sites in the Trace Information Area (14)

The following information is displayed for each site in the trace information area: muscle, sweep, sensitivity and site name. Information common for all sites is written at the top of the trace information area.

You can work with the sites using the ***New Site*** button, the ***Mouse*** or the ***Joysticks/Arrow Keys*** on the Control Panel.

Using the *New Site* button

Activating this button selects the next site or set of sites not yet measured. Use this function during recording mode.

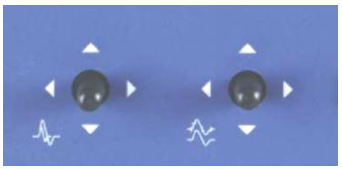
Using the Mouse

- To select a site: click the site (**13**). The site will move up and turn blue, and become active (**12**).
- To deselect a site: click the down arrow of a selected site (**12**). The site will move down and turn pale blue, and become inactive.
- To make an inactive site active: click the pale blue inactive site. The site turns blue, and becomes active.

Using the Joysticks, or the Arrow Keys

The joystick or the arrow keys marked with ∿ can be used while adjusting the markers**.** They select / activate the sites one by one:

- △ (up) and ▽ (down) selects next single site.
- ▷ (right) and ◁ (left) selects marker.



Keypoint Workstation

Keypoint Portable / Keypoint 4

Keypoint G4 / /Keypoint Focus

Other functions of the Control Panel (general to all test types):

The joystick or the arrow keys marked with , modify the sweep and the sensitivity affecting all sites with the same signal type as the currently active sites.

Sweep Speed:

- To increase the sweep speed: click ▷ (right).
- To decrease the sweep speed: click ◁ (left).

Sensitivity Level:

- To increase the sensitivity level: click △ (up).
- To decrease the sensitivity level: click ▽ (down).

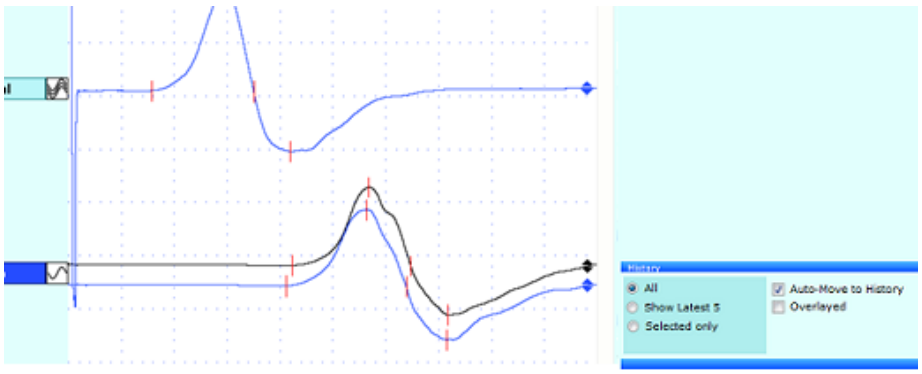
Saving the Recorded Traces

Recorded traces are saved either automatically by selecting the ***Auto-Move to History*** function in the History Window (**7**) (recommended), or by clicking the ***Move to History*** button manually.

NOTE Each site has its own history.

The History and Acquire Views

- Click the ***show all traces*** button to show all traces for the site.
- Click the ***show selected trace*** button to show the selected trace only.
- The selected trace is marked in blue. A new trace always becomes the selected trace. Double click or right click on specific trace to select another.
- Use the ***History*** panel in the right hand side of the screen for additional display options for the selected Site(s).



Stimulation Settings

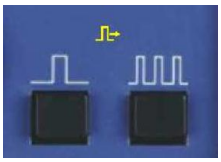
The stimulation settings are predefined, but can be changed through the local test settings. The stimulation settings can be changed using the mouse, or the Control Panel.

Using the mouse

The stimulation settings bar allows you to change the stimulation parameters –i.e. the waveform and the frequency, while the test is running. Click the button, and select the value from the drop-down list.

Using the Control Panel

- Press the ***Single Stimulus*** button to activate the single stimulus pulses, or to stop the repetitive stimulation mode.
- Press the ***Repetitive Stimulation*** button to activate, or to stop the repetitive stimulation mode.



Keypoint

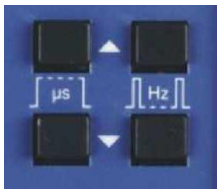
Keypoint G4 / Keypoint Focus

The ***Stimulus Duration*** button sets the stimulus duration as follows:

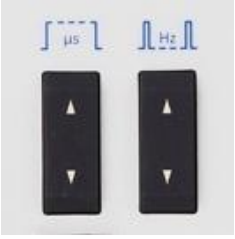
- To increase the duration: click △ (up).
- To decrease the duration: click ▽ (down).

The ***Repetition Rate*** sets the repetition frequency as follows:

- To increase the repetition rate: click △ (up).
- To decrease the repetition rate: click ▽ (down).



Keypoint



Keypoint G4 / Keypoint Focus

Stimulator Overload

WARNING Pay attention to the intensity indicator during the use of the program.

If the intensity indicator starts flashing red, immediately stop the stimulation. See the section on Stimulators in the hardware manual.

Stimulator overload occurs when the current output of the stimulator does not match the desired current.

Check the following possible causes of the stimulator overload:

- One or more electrodes may have fallen off.
- Too high electrode impedance.
- A stimulus set-up requires too much power: high frequency, long pulses, and high intensity.
- Unexpected / abnormal conditions, -e.g. a software or hardware fault, have resulted in an unintended high power stimulus.

Loudspeaker – Control Panel (Keypoint G4 / Keypoint Focus)

Loudspeaker Mute Key / Indicator

- Press the Loudspeaker Mute key to switch between the On and Off functions.

  The yellow light indicates the loudspeaker is muted.

Volume Control Knob / Indicator

- To enable the volume function, press the Volume Control knob.

  The green light volume indicator is on when the volume function is enabled.
- To adjust the volume level, rotate the Volume Control knob while the volume function is enabled.

Markers

Inserting a Marker in the Scope Window

The markers are inserted automatically during the stimulation.

The markers can also be inserted manually:

- Place the cursor over the trace and insert the ***Take off Latency*** marker.
- Select and insert the remaining markers from the pop-up list.

The result tables are updated when a marker is inserted.

Moving a Marker with the Mouse

Click on the marker you want to move, and while pressing down the mouse button, you drag the marker to the wanted position and finish by releasing the button.

- If you drag the marker to the left-hand side, and the marker reaches another marker, both markers are then superimposed.
- If you drag the marker to the right-hand side, and the marker reaches another marker, both markers are then dragged simultaneously.

The result tables are updated when the marker is moved.

Moving a Marker using the Control Panel

Set the wheel to ↔ ***cursor mode*** to move the marker.

The joystick marked with moves the markers, or selects the sites as follows:

- ▷ selects the next marker.
- ◁ selects the previous marker.
- △ selects the Site above.
- ▽ selects the site below.

The wheel moves the selected marker.

Moving a Marker using the Control Panel (Keypoint G4 / Keypoint Focus)

When the Loudspeaker Volume function is enabled , the Trace/Marker Up and Down arrow keys allow you to select the active trace. The Volume Control knob allows you to adjust the loudspeaker volume.

When the Cursor Control Mode function is enabled ↔ , the Trace/Marker Left and Right arrow keys allow you to select the "next" and "previous" marker respectively.

The Trace/Marker Up and Down arrow keys allow you to select the site above and below respectively. With the Cursor Control knob, you can move the selected marker.

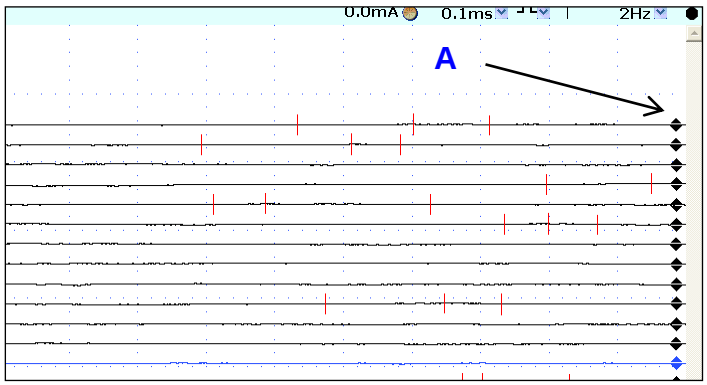
Removing a Marker from the Scope Window

- Dragging a marker beyond the end of the sweep removes the marker.

  The marker is re-entered in the marker pop-up list.

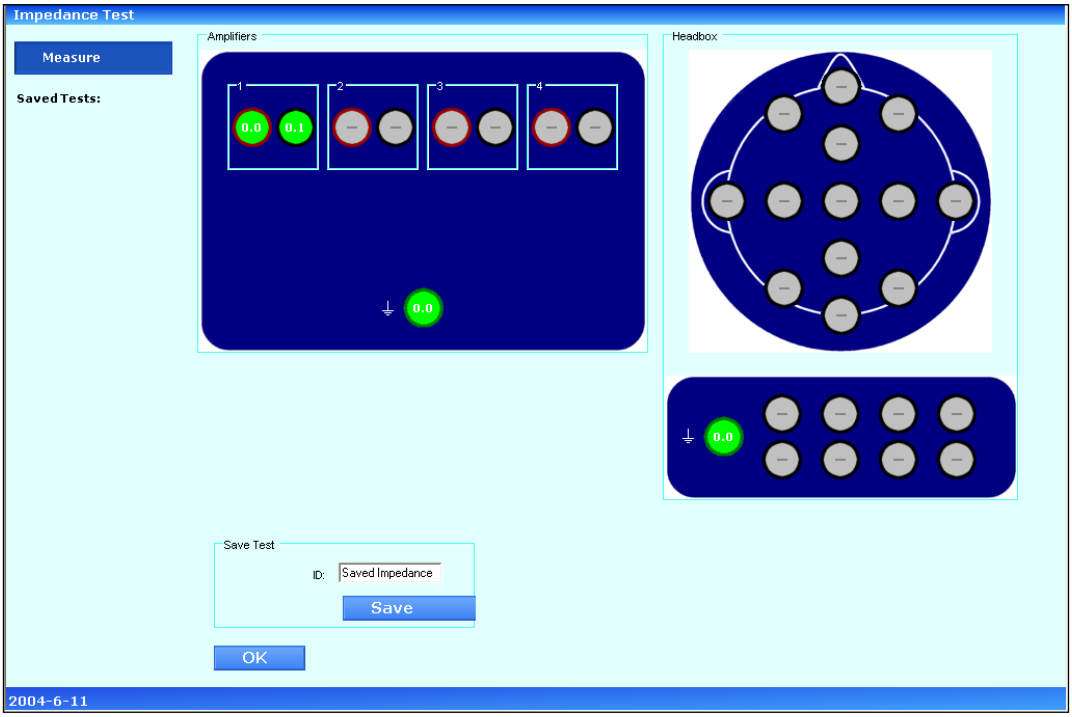
Moving a Trace

In the scope window, the traces can be moved up and down for adjustment of the vertical offset.



- Click on the ***trace handle*** (**A**), at the right end of the curve, while pressing and holding down the mouse button. The trace changes its color, and the hand pointer indicates that you can now move the trace. Finish by releasing the button at the wanted destination.

Impedance



Starting the Impedance Test

- Click on Impedance in the right-click menu or Press ***Ω*** on the Amplifier box (not available for 3-channel Amplifiers). The impedance test starts on all the connected electrodes.
- Adjust the electrodes until an acceptable level is reached (defined in the Settings window). The colored circles will change from red to green. On the Amplifier box, this is indicated by all LEDs being off and high impedance by constantly lit LEDs.

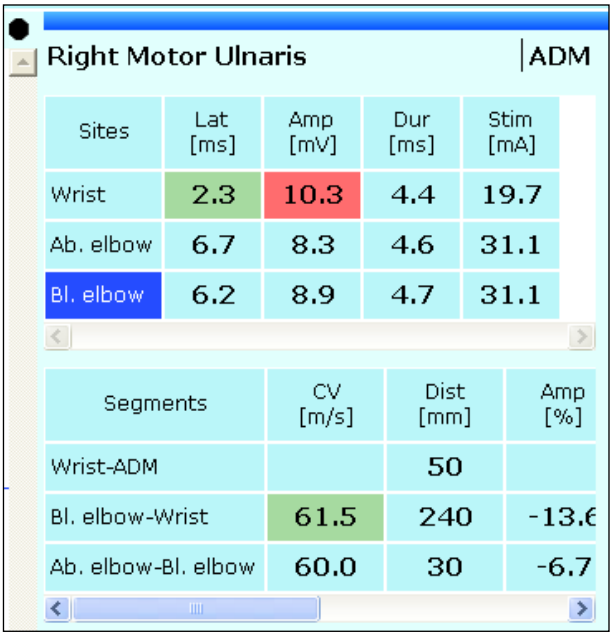
Saving the Impedance Test

- Click ***Save*** to add the impedance test to the ***Saved*** list.

Stopping the Impedance Test

- Click ***OK*** to stop the impedance test.

Working with the Result Tables



Right Motor Ulnaris | ADM

Sites	Lat [ms]	Amp [mV]	Dur [ms]	Stim [mA]
Wrist	2.3	10.3	4.4	19.7
Ab. elbow	6.7	8.3	4.6	31.1
Bl. elbow	6.2	8.9	4.7	31.1

Segments	CV [m/s]	Dist [mm]	Amp [%]
Wrist-ADM		50	
Bl. elbow-Wrist	61.5	240	-13.6
Ab. elbow-Bl. elbow	60.0	30	-6.7

In the motor conduction test, there may be one or two tables according to the anatomy chosen.

The tables can be of ***Site*** and ***Segment*** types.

Until the markers are placed, the tables do not contain any values. The tables reflect the results of the stimulation.

- The distance and the temperature can be edited manually, if they have been selected: click in the field in the table, and enter the value.
- If the result field has a corresponding reference limit, the result will be colored as follows:
  - ***Abnormal:*** red background.
  - ***Normal:*** light green background
  - ***Supernormal*** (i.e. normal, but outside the limit): dark green background.
- Clicking on a site name in the table activates the corresponding trace in the scope window.

Optional Windows

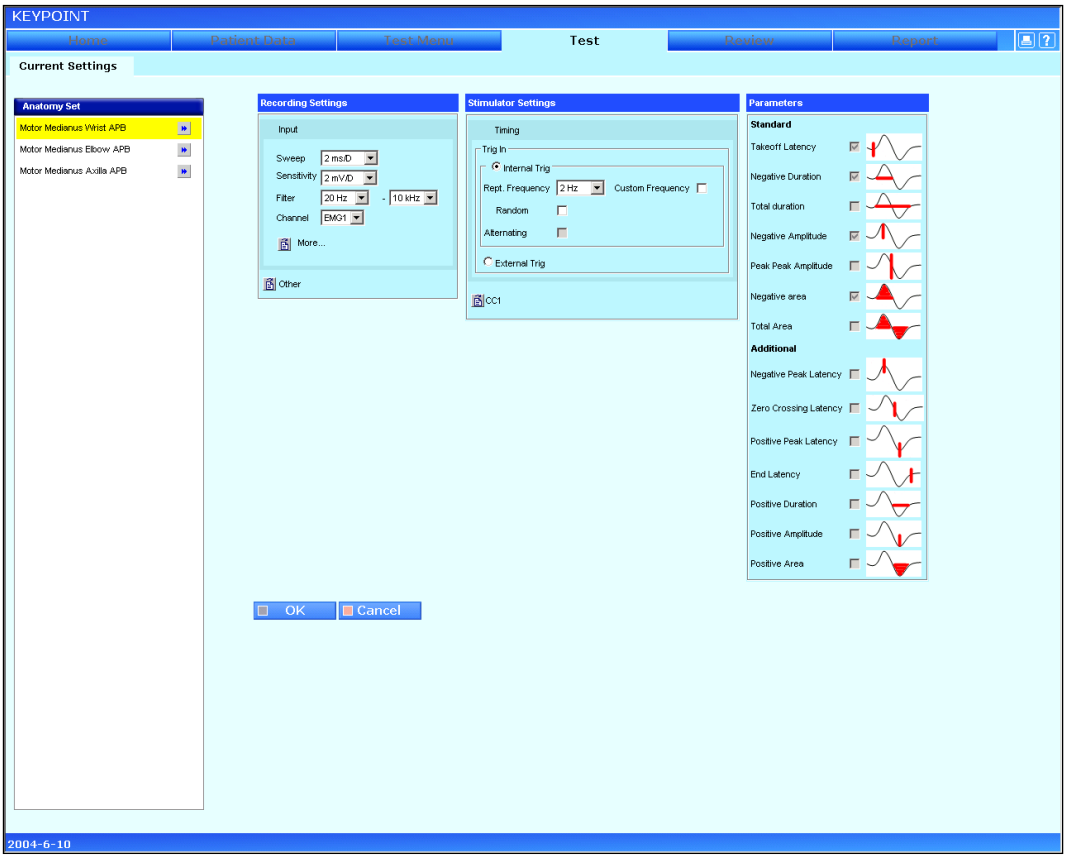
If you click the *right* mouse button *outside* the traces within the scope window a set of optional windows are available. The Activity Window, a Plot Window and a Note Window.

Nerve Conduction Data Page

Activating the ***NC Data*** button takes you to a summary page. Clicking this button again, or closing the summary page window, takes you back to the Test page.

Settings and Design

The Settings Page



From the ***Current Settings*** page, the recording and stimulation parameters of the test can be adjusted.

- Click on the anatomy set to view the settings of a site.
- To open the site settings of a specific trace from the scope window, right-click on the trace, and select settings.
- After adjusting the settings, click ***OK*** to save the settings, or click ***Cancel*** to quit the settings page without saving the changes.

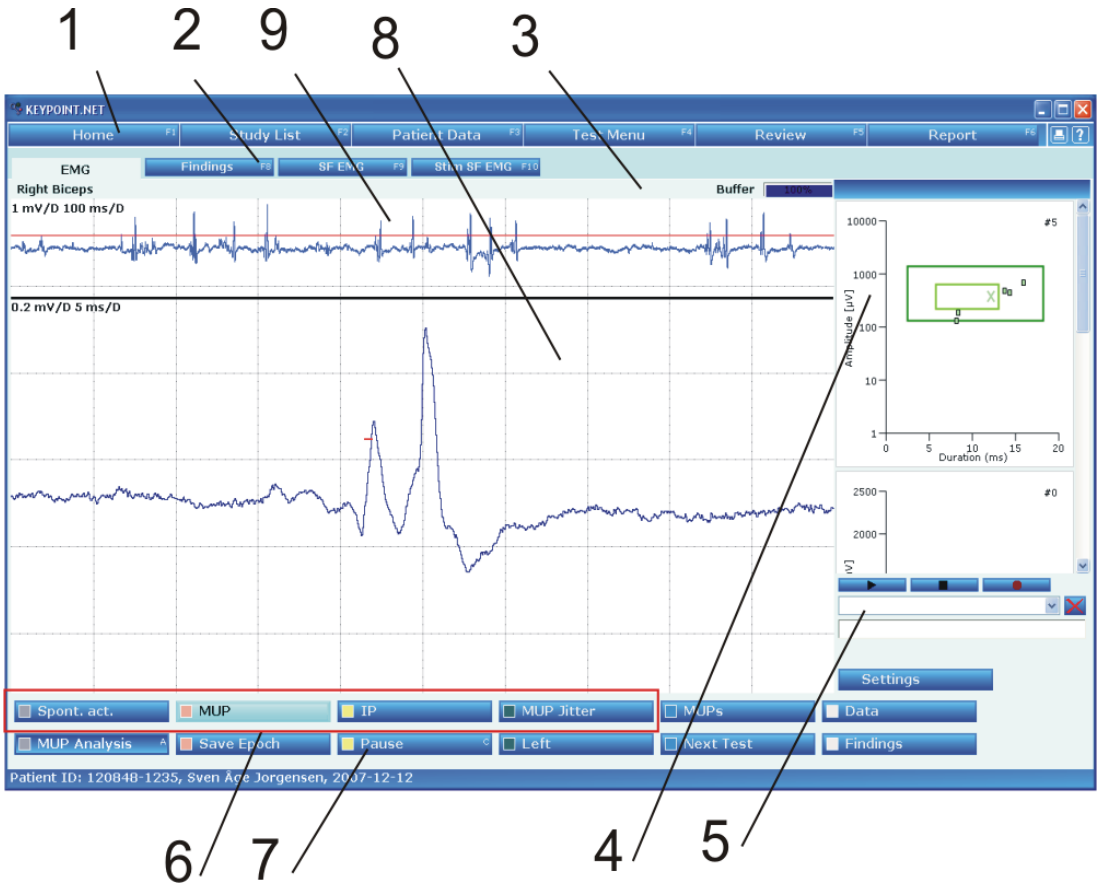
The customized settings carried out through Settings and, if available, Design pages, are only valid for that specific test and patient. New tests will have the default test settings.

Test Example 2: EMG

Introduction

Test Example 2 describes the functions available for the Electromyography (EMG) test.

Overview – Acquire View



1 Top navigation bar with tabs

2 Test navigation tabs

3 Scope Top bar

4 Plot windows

5 Tape recorder

6 View selection buttons

7 Color marked buttons (for use with the dedicated keys)

8 Main scope

9 Overview scope with sweep and sensibility values

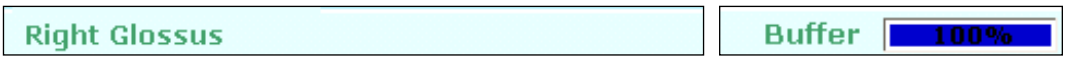
Notes on Functions and Views

View selections (6)

- A set of preconfigured ***Views*** determine the functionality of the program when it is in acquire mode. The ***Buttons*** –see red framed buttons in the figure above– control the currently selected view. When selected, the button turns light blue.

- A ***View*** is configured with a number of parameters like the display of overview curves, gain and sweep speed, and analysis buttons available.
- The default setup for the ***Views*** can be modified through ***Options*** from the Home page.

Scope Top Bar (3)



The scope top bar displays the name and the side of the muscle and the status of the buffer. The buffer collects data during a number of seconds, e.g. 5sec. The number of seconds is user-definable.

Trigger Controls

Working in the main scope window (**8**), two triggers controls options exist and can be activated either by way of the mouse, or by using the joysticks on the control panel:

Using the Mouse

- The ***trigger*** can be added to the trace by clicking anywhere in the main scope (**9**).

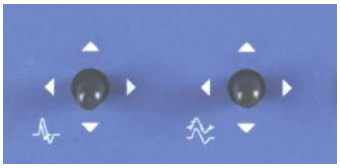
  You can drag the trigger to any position in the scope: move the mouse pointer to the trigger, and then hold down the left mouse button, while moving it to the wanted position. Finish by releasing the mouse button.

  Remove the trigger from the scope by dragging it outside the scope; the main scope window then turns into free running mode.

Using the Joysticks on the Control Panel

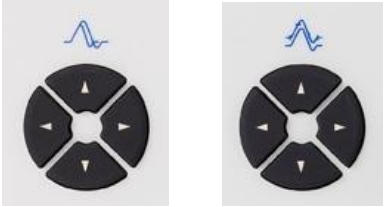
The dedicated joystick marked with ∿ modifies the ***trigger delay***:

- To increase the trig delay: click ▷ (right).
- To decrease the trig delay: click ◁ (left).



Keypoint Workstation

Keypoint Portable and Keypoint 4



Keypoint G4 / Keypoint Focus

A* Keypoint Portable and Keypoint 4 only

Trigger Level

Press the wheel mode button (**A/A***) to switch to ***Cursor Mode***. The ↔ indicator turns yellow.

- The intensity wheel now adjusts the trigger level.

Trigger Level (Keypoint G4 / Keypoint Focus)

Press the Cursor Control knob to switch to ***Cursor Mode***. When the Cursor Mode is enabled, the green light indicator is on ● ↔.

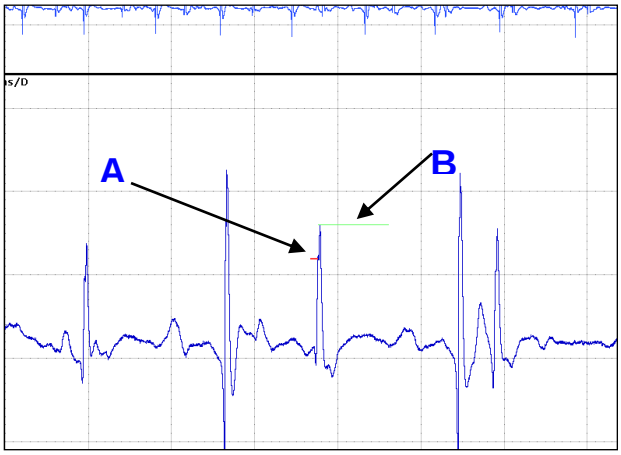
- The Cursor Control knob allows you to adjust the level of the trigger.

Trigger Slope

To select/deselect the trigger slope: open the right-click menu in the scope, and select "Trig Slope".

- If the trigger slope is selected, the signal is triggered on a down going signal.
- If the trigger slope is deselected, the signal triggers on an up going signal.

Window Trigger



- The ***window trigger*** consists of the normal trigger (**A**) and the window indicator (**B**), and can be selected in the right-click menu. Please, see the figure above.

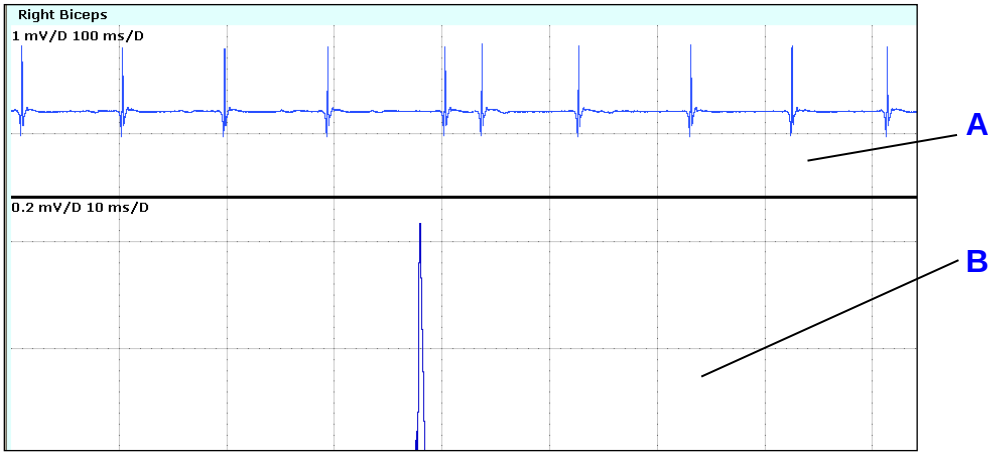
  You can change the length of the window trigger by dragging it from the right side of the indicator.

  The window indicator can be moved up and down independently of the trigger by dragging the mouse.

Split Screen View

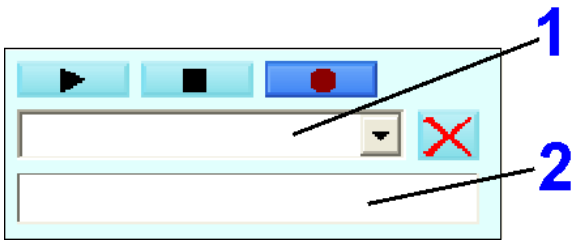
To split the window into two viewing panes, click the *right* mouse button in the scope, and select "split" from the menu, and the window splits into an ***overview scope*** and a ***main scope*** as shown in the figure below:

- ***The overview scope* (A):** when a trigger is applied to the main scope, a thin red line in the overview scope indicates the amplitude level.
- ***The main scope* (B):** the trigger is set in the main scope in which curves in free run are shown.



The Digital Tape Recorder

The digital tape recorder is used to record live tests, or to play existing recordings. All recordings are hard disk recordings.



1 Recording list

2 Status bar shows the progress in time during the playback.

Play or ***Pause*** a recording.

Stop recording or playing

Record signals live.

Delete a recording selected in the recording list.

Exporting a Signal

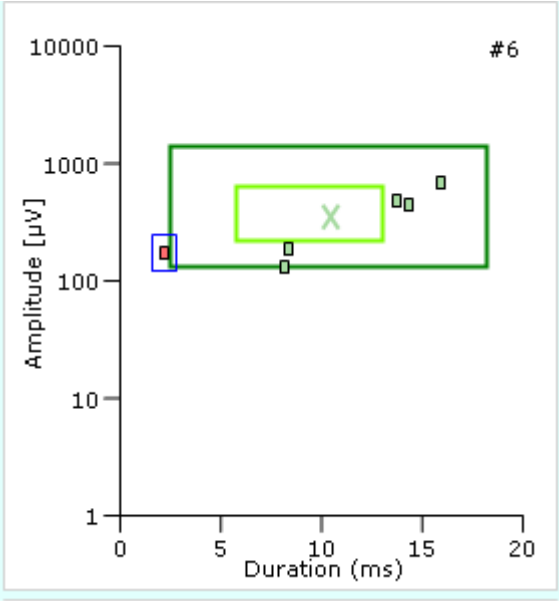
Right-click on the status bar to export the recording to either wave or text format.

Plot Windows (4) (optional)

Depending on the Test configuration, this area shows different result plot windows: ***Inter-discharge interval*** (***IDI***), ***Motor Unit Potential*** (***MUP***), ***Turns/Amplitude***, ***Number of Small Segments*** (***NSS***), ***Envelope,*** and ***Peak Ratio***.

Note that the plot windows do not contain any values until an analysis has been conducted.

Example: The Motor Unit Potential Result Plot Window (optional)



The MUP result plot window displays the MUPs, which are found, when analyzing. The number of accepted MUPs is displayed in the upper-right corner of the window.

- The square represents the MUP. To activate the MUPs Edit, which includes detailed data for the MUP, amplitude and duration, simply click on the square.
- The mean amplitude/duration is displayed as an X (the color codes are similar to the MUP).

If the MUP has a corresponding reference limit, the result will be colored as follows:

- ***Abnormal:*** a red square.
- ***Normal:*** a green square.

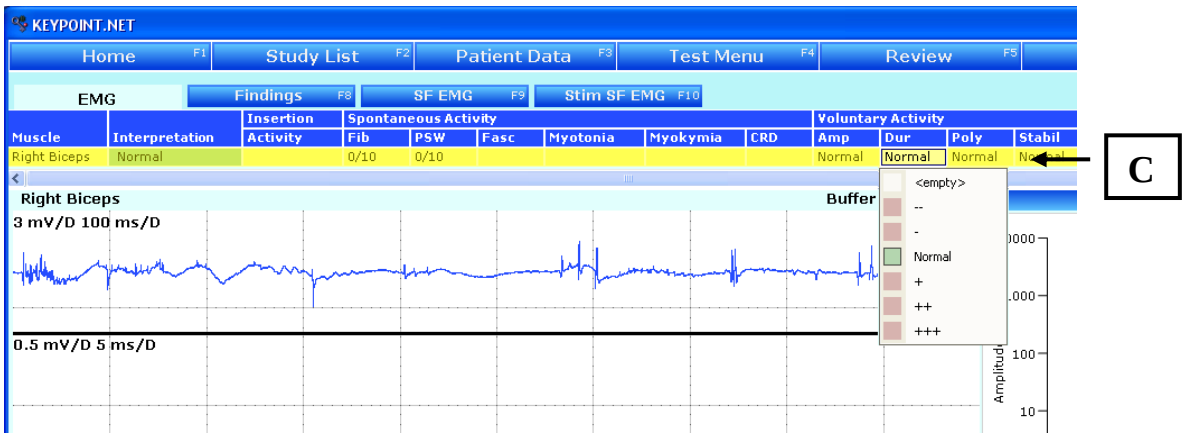
If the reference values are present, there may be one box for the single MUPs and one box for the mean MUPs.

MUPs and amplitude/duration without a corresponding reference limit are black.

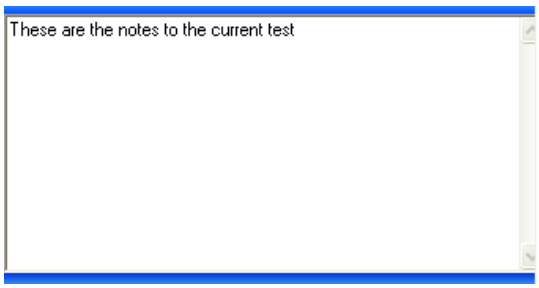
The normal area is displayed as a transparent box with a green border.

Findings

Use the ***Findings*** button to go to the Findings page. This page allows complex scoring of your findings for all muscles. –Alternatively, activate the ***Findings bar*** **(C)** on the acquire view to allow on-line scoring for the selected muscle. The Findings bar can be activated by right-clicking in the curve window or by selecting a view where it is predefined as active. In both cases, scoring is changed by clicking in the table cells. Note that you can change the score with a single click by keeping the right mouse button pressed while moving the mouse pointer to the wanted score.



Note Window



To display the ***note window***, select it in the right-click menu.

In the ***note window*** you can type in comments related to the current test and side.

Click in the note window, and the blinking cursor indicates that you can now type in the text.

The functions of undo, cut, copy, paste, delete, or select are available at the *right* click of the mouse button in the note window.

If a note window is present, but not displayed, a small icon is visible next to the name of the anatomy in the top left part of the screen.

The Notes field on the Patient Data page displays all the entered test notes.

MUP Analysis (optional)

When clicking the ***MUP Analysis button***, the application analyzes the data from the MUP session, searching for possible MUPs. Each ***unique, repeated*** MUP is identified.

The analysis identifies and averages individual discharges from the same motor unit. The action after analysis is predefined and depends on the Test configuration.

- Types of actions if MUPs are found:
  - Stay in Current View (auto accept).
  - Go to MUP Edit, if any MUPs are found.
  - Go to MUP Review View.
  - Go to MUPs.

- Types of actions, if no MUPs are found:
  - Stay in the current view (continue).
  - Go to the MUP review view.

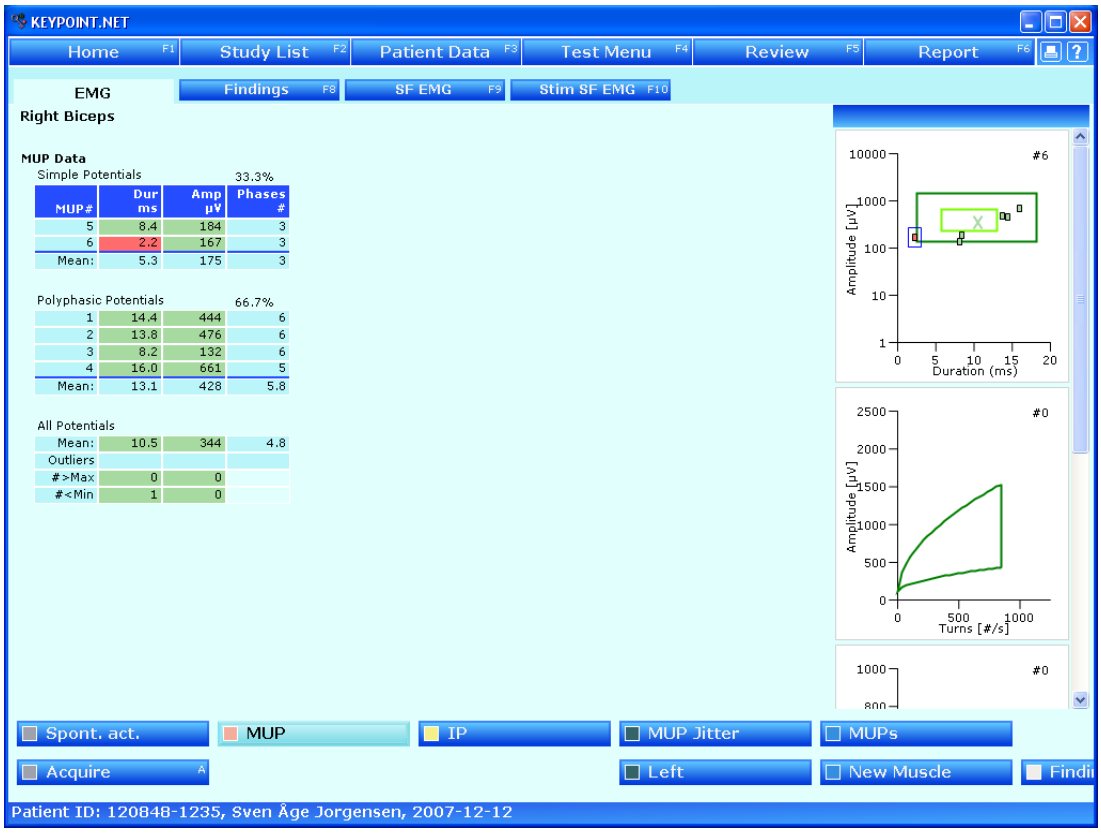
IP Analysis (optional)

When clicking the ***IP analysis button***, the application analyses the data from the IP (Interference Pattern) session, calculates the turns, the number of small segments, NSS, and other IP parameters.

Motor Unit Potentials (MUP) (optional)

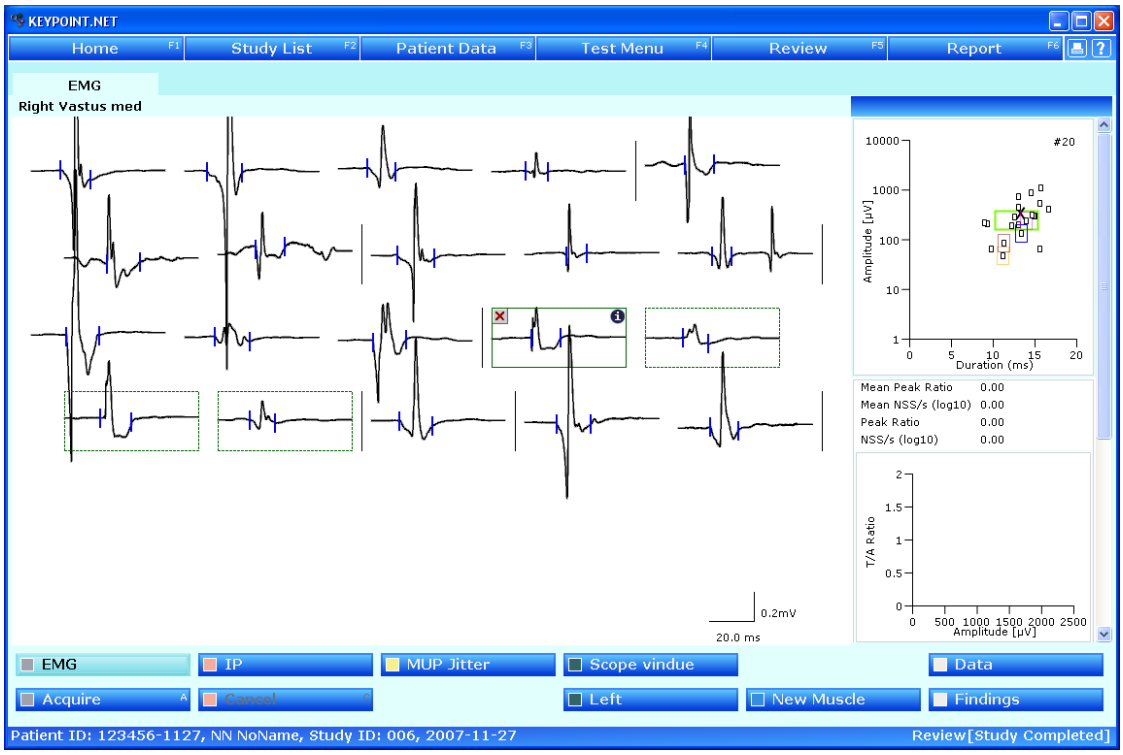
MUP – Data Page

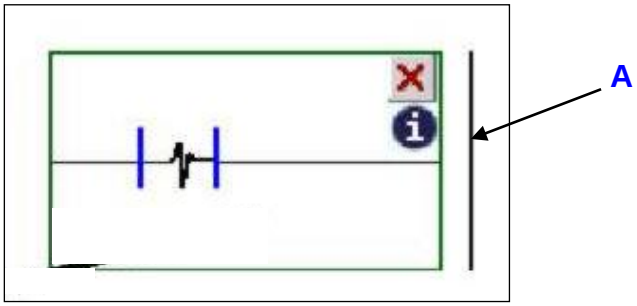
The ***Data*** page displays a table of the motor unit potentials with detailed parameters as shown in the figure below.



MUP Page

In the ***MUP*** page, you can see the recordings / epochs. The MUPs are broken down into single units by a vertical line as shown in the figure below.





A zoom-in on one of the MUPs.

The following functions become available when you place the cursor over a MUP that you would like a closer look at:

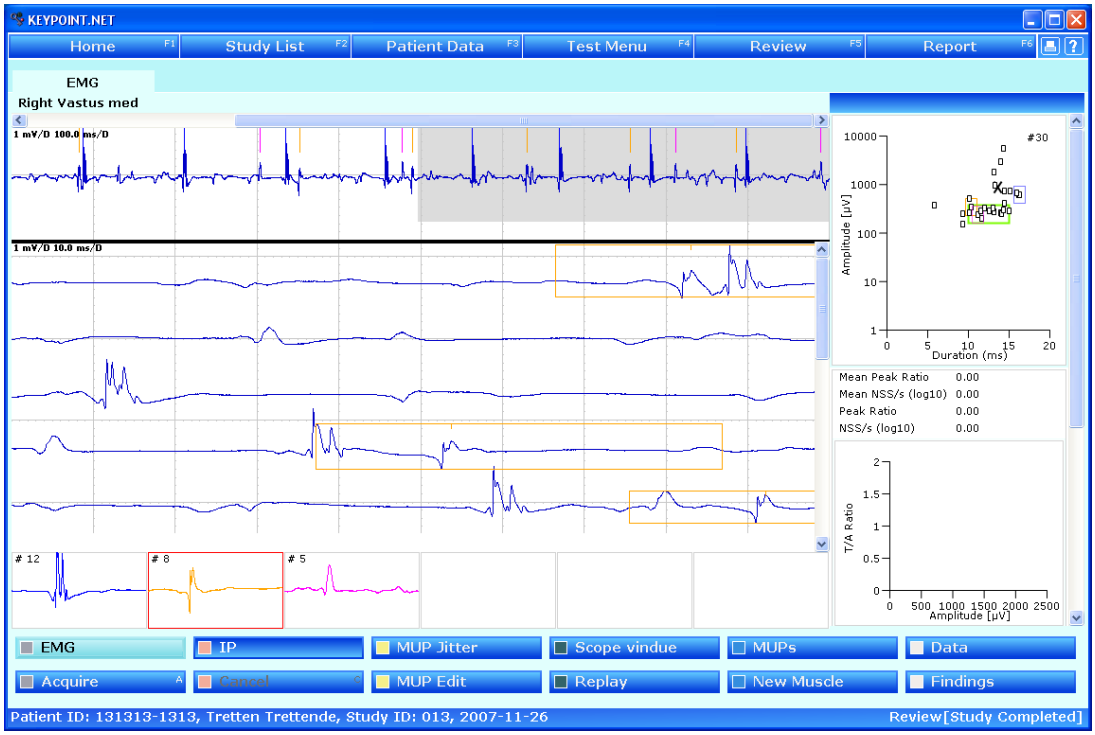
- Click ✕ to delete the MUP. The MUP is then marked with a red cross.
- Click ℹ to open the MUP in the MUP Edit view.
- Click and drag a marker to adjust. This is performed in a zoom window.
- The vertical dividing line (**A**) indicates the end of a MUP recording. Click at a marker to show this MUP in adjust mode.

MUP – Review Page

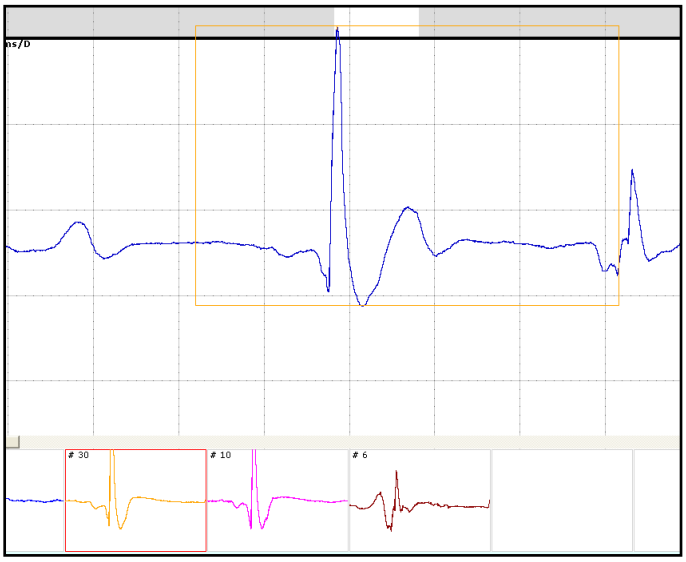
In the ***MUP Review*** page, you can see the data from each recording / epoch.

From this page you can also group MUPs manually, if the automatic MUP detection does not work on a specific signal. You can access this view through the MUP Edit view.

Note that the ***Overview Scope*** is shown at the top; the ***Main Scope*** in the middle; and boxes with ***Template*** curves at the bottom.



Left click to select a template. The firing pattern is shown in the overview scope by small lines. In the main scope, a box marks the enclosed single MUP.



Locating a MUP

When moving the mouse in the scope, a box will appear around the detected MUP. A small tab in the middle of the box indicates the presence of a MUP.

Adding a MUP to a Template

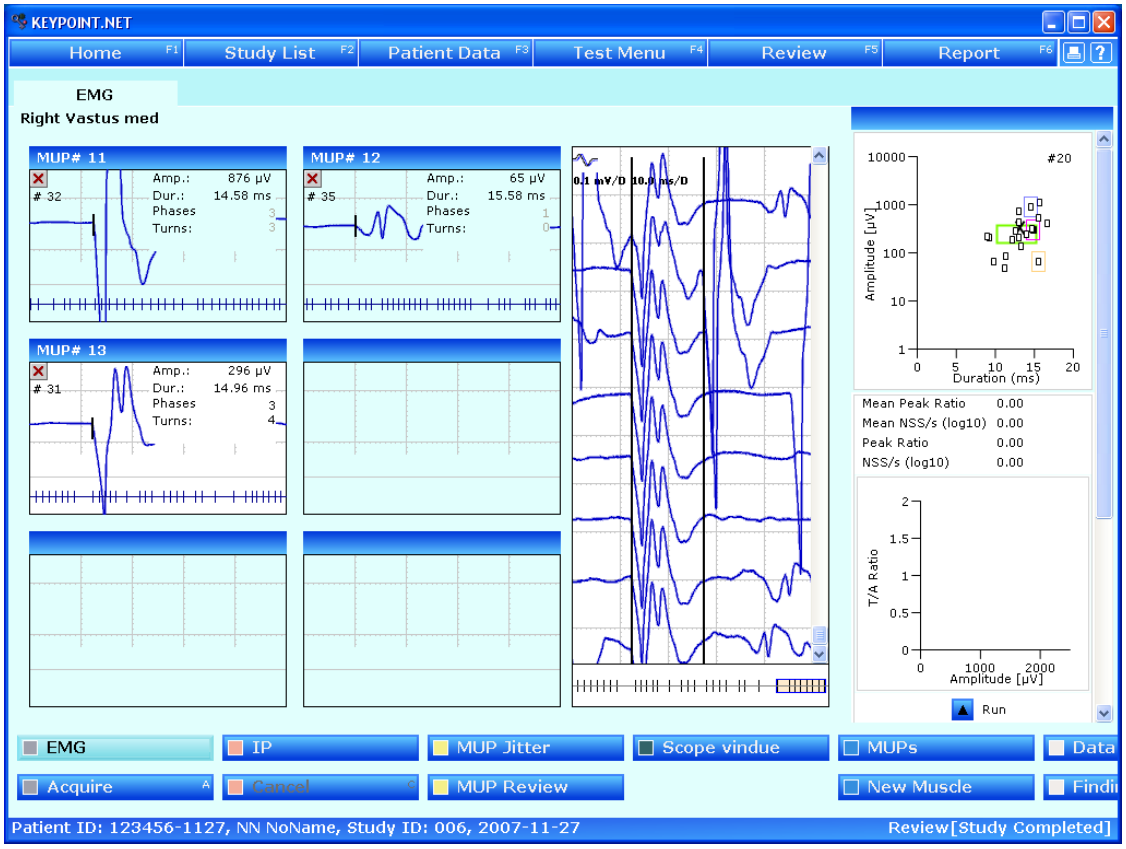
Click on a MUP and the program selects the MUP closest to the small tab in the box. The box is automatically moved to the peak, and the MUP is added to the selected template box.

Right-click inside the box to open a menu for including, excluding or auto searching similar MUP discharges for the MUP template.

Removing a MUP from a Template

Click on an already selected MUP to remove the MUP from the template box.

MUP – Edit Page



Template Area

The template comprises the following information:

- The ***amplitude*** and the ***duration*** are displayed in green, if normal; in red, if abnormal, and in black, if there are no reference values available.
- ***Phases*** (can be edited).
- ***Turns*** (can be edited).
- ***The number of discharges.***

At the bottom of each box the firing pattern is shown by small “sticks”.

Editing the Template

Each active template displays a MUP with two markers. The markers can be moved. The change is reflected in the Raster Window.

Averaging and Merging two Templates

In the bottom of the active template there is a firing pattern. To perform an average of the two templates' signals and a merge of the firing pattern, simply click and drag the "sticks" pattern to another template.

Deleting a Template

Click ✕ to delete the template. The template is marked with a red cross.

Raster Window

The raster window displays all the single MUPs from the selected template.

Right click in the window to turn the feature of superimpose on and off.

Clicking a signal in the raster window toggles the signal on and off.

Other Functions

Backup

CAUTION It is strongly recommended to back up regularly.
To back up the database, use the Admin Tool program.
To back up the directory containing archived studies, use the Windows Backup Tool.

Screen Print

Screen print is activated by left-clicking the print icon in the top right corner or by pressing ***<ctrl-print screen>*** on the PC keyboard. The printer to be used and other settings can be changed by right-clicking the print icon.

Demo Mode

For demonstration and teaching purposes the program can run in Demo mode where simulated signals are shown. This mode can be enabled by clicking Options, and then System from the Home Page.

In Demo mode use ***<ctrl s>*** to simulate a single stimulation and ***<ctrl r>*** to toggle repetitive stimulation on and off.

Right-Click

Note that if a function −e.g. button− seems to be missing, it is recommended to try right-clicking at various elements on the screen. A menu with one or more options will be displayed.

Software Warning Messages

During use, the following software messages may occur:

Intensity > 120 dB !! Increase ?
To go beyond 120 dB, you must answer yes.

WARNING
New reference values are required, when changing the algorithm.
If you use the reference value system, the values must reflect the algorithm you select.

WARNING
New settings require new reference values.
If you use the reference value system, the values must reflect the settings you select.

Overloaded – Reduce Sensitivity!!
Saturation has occurred, if this is a consistent error message, it is advisable to reduce the input sensitivity, since shape parameters can be heavily influenced by saturation.

Gain Errors in Amplifiers, max. error.
Gain in amplifiers, as tested during power up, exceeds the accepted calibration limits. Exit the Keypoint·NET program, and verify the cable connection. Re-calibration may be necessary.

Current Stimulator may be malfunctioning.
Problem occurred during the power up test of the stimulator. Exit the Keypoint.NET program, and verify the cable connection. Do not use the current stimulator, when this message is shown, as the stimulator might be defective.

No Hardware detected.
Communication between PC and Hardware cannot be established. Exit the Keypoint.NET program, and check:
- the USB cable connection between the computer unit and the Connection Panel (internal);
-the power cable connection to the Connection Panel (internal).

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